SUPPLEMENTARY MATERIAL

Glucose-Lowering and the Risk of Cardiovascular Events with Novel Antidiabetic Therapies: A Systematic Review and Additive-effects Network Meta-Analysis

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Table S1. Baseline characteristics of patients in the 126 randomized controlled trials (143 study arms)

RCT	Active arm	Control arm	Follow-up time (years, mean)	Sample size	Weight	Exposure (patient-years)	Events patient-years (Active arm)	Events patient-years (Control arm)	Women (%, mean)	Age (years, mean)	Time since DM diagnosis (years)	HbA1c at baseline (%, mean±SD)
ACCORD (2008)(1)	Sulfonylurea+Insulin	Sulfonylurea+Insulin	3.50	10251	4,97%	35879	19.61	20.69	38.7	62.2	10	8.30 ± 0.90
ADVANCE (2008)(2)	Sulfonylurea+Insulin	Sulfonylurea+Insulin	5.00	11140	7,72%	55700	20	21.19	42.4	66.00	8.00	7.52 ± 1.40
AWARD-3(3)	GLP1A	Metformin	0.96	807	0,11%	775	3.87	0	55.00	55.00	3.00	7.60 ± 0.80
AWARD-5(4)	GLP1A	DPP4i	1.90	921	0,24%	1750	5.21	6.68	52.00	54.00	7.00	8.10 ± 1.10
AWARD-9(5)	GLP1A	Placebo	0.53	300	0,02%	159	25.16	0	41.30	60.60	13.30	8.30 ± 0.80
Abdul-ghani (2017) (6)	GLP1A+TZD	Insulin	1.00	231	0,03%	231	0	46.30	63.00	52.00	10.90	10 ± 0.50
Arechavaleta (2010) (7)	DPP4i	Sulfonylurea	0.58	1035	0,08%	600	0	6.64	53.80	56.20	6.70	7.50 ± 0.80
Arjona (2013) (8)	DPP4i	Sulfonylurea	1.00	423	0,06%	423	23.70	23.58	45.10	64.30	10.10	7.80 ± 0.70
Bailey (2013) (9)	SGLT2i	Placebo	2.00	546	0,15%	1092	7.33	14.60	45.00	53.70	5.80	8.11 ± 0.96
Bailey (2015) (10)	SGLT2i	Placebo	2.00	274	0,08%	548	5.03	0	58.70	52.70	2.10	7.84 ± 0.87
Barnett (2013) (11)	DPP4i	Placebo	0.96	455	0,06%	437	6.85	6.90	55.00	57.30	12.20	8.60 ± 0.90
Barnett (2014) (12)	SGLT2i	Placebo	1.00	374	0,05%	374	5.35	5.35	43.30	65.10		8.09 ± 0.80
Bode (2015) (13)	SGLT2i	Placebo	1.96	714	0,19%	1399	9.63	6.46	39.70	63.20	11.40	7.80 ± 0.80

Bolli (2009) (14)	DPP4i	SGLT2i	0.96	576	0,08%	553	7.06	11.12	35.90	57.00	6.40	8.40	±	0.90
Bosi (2007) (15)	DPP4i	Placebo	0.46	544	0,03%	250	6.01	0	46.90	54.50	6.20	8.30	±	0.90
Bosi (2011) (16)	DPP4i	SGLT2i	0.96	803	0,11%	771	5.16	7.83	48.90	55.90	6.90	8.10	±	0.83
Buse (2011) (17)	GLP1A+Insulin	Insulin+Placebo	0.57	259	0,02%	148	0	14.38	36.00	59.00	12.00	8.50	±	0.96
CANVAS(18)	SGLT2i	Placebo	3.50	10142	4,92%	35497	26.62	30.96	36.70	63.40	13.70	8.20	±	0.90
CARMELINA(19)	DPP4i	Placebo	2.20	6979	2,13%	15354	56.46	54.78	35.70	65.60	14.50	8.00	±	1.00
CAROLINA(20)	DPP4i	Sulfonylurea	6.30	6033	5,27%	38008	18.69	19.09	40.80	64.20	6.20	7.20	±	0.60
CONFIDENCE (1) (21)	GLP1A	Insulin	0.92	224	0,03%	206	9.88	9.53	38.60	51.00		8.20		
CONFIDENCE (2) (21)	GLP1A	Pioglitazone	0.92	228	0,03%	210	9.88	9.21	44.90	50		8.00		
Cefalu (2015) (22)	SGLT2i	Placebo	0.96	914	0,12%	877	13.74	6.81	31.40	63.00	12.30	8.08	±	0.80
Chacra (2011) (23)	DPP4i	Placebo	1.40	769	0,15%	1077	11.38	10.70	53.90	55.10	6.80	8.40	±	0.90
Charpentier (2009) (24)	Pioglitazone	Placebo	0.58	289	0,02%	168	12.08	0	35.00	60.20	12.50	8.10	±	0.70
Chen (2018) (25)	GLP1A	Sulfonylurea	0.50	720	0,05%	360	4.18	0	46.30	52.00	3.80	7.90	±	1.01
Davies (2021) (26)	GLP1A	Placebo	1.25	1210	0,21%	1512	8.8	2.4	49.95	55.25	8.1	8.10	±	0.80
DECLARE(27)	SGLT2i	Placebo	4.20	17160	9,99%	72072	20.97	22.29	37.90	64.00	10	8.30	±	1.20
DUAL IV(28)	GLP1A+Insulin	Placebo	0.50	435	0,03%	218	13.84	0	50	59.40	9.30	7.90	±	0.60

DURATION-2 (1) (29)	GLP1A	DPP4i	0.50	326	0,02%	163	0	12.05	48.00	52.00	5.00	8.50	±	1.20
DURATION-2 (2) (29)	GLP1A	Pioglitazone	0.50	325	0,02%	163	0	36.36	52.00	53.00	6.00	8.50	±	1.10
DURATION-4 (1) (30)	GLP1A	Metformin	0.50	494	0,03%	247	0	8.13	37.40	54.00	2.60	8.60	±	1.20
DURATION-4 (2) (30)	GLP1A	TZD	0.50	411	0,03%	206	0	12.27	40.50	55.00	2.70	8.50	±	1.20
DURATION-7(31)	GLP1A+Insulin	Insulin+Placebo	0.53	461	0,03%	244	0	8.20	53.50	57.60	11.10	8.53	±	0.92
DURATION-8 (1)(32)	GLP1A	SGLT2i	0.96	457	0,06%	439	22.94	18.12	52.20	54.50	7.10	9.30	±	1.00
DURATION-8 (2) (32)	GLP1A+SGLT2i	SGLT2i	0.96	458	0,06%	440	9.14	18.12	52.20	54.50	7.10	9.30	±	1.00
DURATION-8 (3) (32)	GLP1A+SGLT2i	GLP1A	0.96	455	0,06%	437	9.14	22.94	48.90	54.20	7.40	9.30	±	1.10
DeFronzo (2009) (33)	DPP4i	Placebo	0.46	743	0,05%	342	0	12.14	46.40	54.80	6.70	8.10	±	0.90
Del Prato (2014) (34)	DPP4i	Sulfonylurea	1.92	2639	0,70%	5067	4.13	6.56	49.50	55.40	5.50	7.60	±	0.62
Del Prato (2015) (34)	SGLT2i	Sulfonylurea	4.00	801	0,44%	3204	0.63	2.49	45.10	58.60	6.60	7.74	±	0.89
Dormandy (2005) (35)	Pioglitazone	Placebo	2.88	5238	2,09%	15059	44.33	51.12	34.00	61.60	8.00	7.90		
ELIXA(36)	GLP1A	Placebo	2.08	6068	1,75%	12621	62.91	61.64	30.90	60.60	9.40	7.60	±	1.30
EMPA-REG(37)	SGLT2i	Placebo	3.10	7020	3,02%	21762	33.72	38.99	28.00	63.20		8.08	±	0.80
EXAMINE(38)	DPP4i	Placebo	1.50	5380	1,12%	8070	75.28	78.64	32.00	60.70	9.18	8.00	±	1.10
EXSCEL (39)	GLP1A	Placebo	3.20	14752	6,54%	47206	35.64	38.24	38.00	62.00	12.00	8.00		

Ferdinand (2014) (40)	GLP1A	Placebo	0.50	755	0,05%	378	3.96	8.00	47.60	56.40	8.40	7.90	±	0.80
Ferrannini (2009) (41)	Sulfonylurea	DPP4i	1.00	2789	0,39%	2789	15.79	8.60	47.20		5.71	7.31	±	0.64
GET-GOAL(42)	GLP1A	Placebo	1.40	482	0,09%	675	6.65	4.46	55.00	58.20	6.20	8.00	±	0.80
Gallwitz (2012) (43)	DPP4i	Sulfonylurea	2.00	1551	0,43%	3102	7.09	14.84	39.00	59.80		7.70	±	0.90
Gantz (2017) (44)	DPP4i	Placebo	1.80	4202	1,05%	7564	30.16	30.13	29.30	63.60	12.10	8.00	±	0.90
GetGoal-Duo-1(45)	GLP1A	Placebo	0.46	446	0,03%	205	9.75	19.50	51.00	56.00	9.60	7.60	±	0.50
GetGoal-L(46)	GLP1A	Placebo	0.50	495	0,03%	248	18.29	35.93	51.00	57.00	12.40	8.40	±	0.80
GetGoal-L-Asia(47)	GLP1A	Placebo	0.46	311	0,02%	143	28.23	0	49.00	58.00	14.10	8.52	±	0.78
GetGoal-M-Asia(48)	GLP1A	Placebo	0.46	390	0,02%	179	22.18	11.21	53.10	55.10	6.80	7.85	±	0.71
GetGoal-O (49)	GLP1A	Placebo	0.46	350	0,02%	161	0	37.48	48.30	74.40	14.60	8.05	±	0.70
GetGoal-S(50)	GLP1A	Placebo	0.46	859	0,05%	395	30.30	30.51	47.40	57.80	9.80	8.20	±	0.80
Giles (2008) (51)	Pioglitazone	Sulfonylurea	0.50	518	0,04%	259	68.70	62.50	23.00	63.40	11.71	8.95		
Giles (2010) (52)	Pioglitazone	Sulfonylurea	1.00	300	0,04%	300	0	13.42	44.00	64.00		8.30		
Goke (2008) (53)	DPP4i	Metformin	2.00	463	0,13%	926	1.64	0	39.00	54.08	2.20	8.80	±	0.10
Goke (2013) (54)	DPP4i	Sulfonylurea	1.90	858	0,23%	1630	3.69	7.34	46.00	57.60	5.40	7.65	±	0.04
HARMONY(55)	GLP1A	Placebo	1.50	9463	1,97%	14195	47.63	60.30	31.00	64.20	14.20	8.72	±	1.50

HARMONY-1(56)	GLP1A	Placebo	2.90	301	0,12%	873	4.60	2.28	41.70	54.90	7.90	8.10	±	0.90
HARMONY-3 (1) (57)	GLP1A	Sulfonylurea	1.90	609	0,16%	1157	12.20	3.43	48.50	54.40	6.00	8.10	±	0.80
HARMONY-3 (2) (57)	GLP1A	DPP4i	1.90	604	0,16%	1148	12.20	3.49	54.00	54.30	5.80	8.10	±	0.80
HARMONY-3 (3) (57)	GLP1A	Placebo	1.90	403	0,11%	766	12.20	15.63	50.50	56.10	6.70	8.20	±	0.90
HARMONY-5 (1) (58)	GLP1A	Pioglitazone	2.90	548	0,22%	1589	1.27	7.47	46.60	55.70	9.20	8.29	±	0.88
HARMONY-5 (2) (58)	GLP1A	Placebo	2.90	386	0,16%	1119	1.27	9.00	39.10	55.70	9.30	8.26	±	0.98
HARMONY-8(59)	GLP1A	DPP4i	0.96	495	0,07%	475	20.92	29.64	47.20	63.50	11.62	8.23	±	0.94
Haering (2015) (60)	SGLT2i	Placebo	1.40	666	0,13%	932	6.48	12.70	50.20	56.90		8.20	±	0.80
Handelsman (2017) (61)	DPP4i	Sulfonylurea	1.00	751	0,10%	751	7.98	5.33	43.70	58.00	7.70	7.40	±	0.70
Henry (2013)-1(62)	DPP4i	Pioglitazone	1.04	751	0,11%	781	10.34	3.40	43.50	51.15	3.76	8.80	±	1.10
Henry (2013)-2(62)	DPP4i	DPP4i+ Pioglitazone	1.04	767	0,11%	798	10.34	6.62	43.50	53.40	3.97	8.80	±	1.10
Henry (2013)-3(62)	DPP4i+TZD	Pioglitazone	1.04	1146	0,17%	1192	6.62	3.40	43.50	51.15	3.76	8.80	±	1.10
Hollander (2011) (63)	DPP4i	Placebo	1.40	565	0,11%	791	9.37	3.88	53.80	54.00	5.10	8.20	±	1.10
Hollander (2019) (64)	SGLT2i	Sulfonylurea	2.00	1305	0,36%	2610	9.77	5.75	48.50	57.90	7.60	7.80	±	0.60
Hong (2013) (65)	Sulfonylurea	Metformin	3.00	304	0,13%	912	72.07	49.15	21.80	62.80	5.60	7.60	±	1.70
IDegLira(66)	GLP1A+Insulin	Insulin	0.50	398	0,03%	199	10.05	20.10	47.00	58.00	11.00	8.80	±	0.70

Jain (2006) (67)	Pioglitazone	Sulfonylurea	1.00	502	0,07%	502	7.97	11.95	43.80	52.10	9.40	9.20	±	1.26
Ji (2021) (68)	GLP1	DPP4	0.55	868	0,07%	477	1.8	0	40.9	63.00	6.30	8.1	±	0.9
Kawamori (2018) (69)	SGLT2i	Placebo	0.96	275	0,04%	264	5.72	11.20	22.60	59.80	8.70	8.36	±	0.74
Kohan (2014) (70)	SGLT2i	Placebo	1.96	252	0,07%	494	15.18	36.44	36.90	67.00	15.70	8.53	±	1.28
Kovacs (2015) (71)	SGLT2i	Placebo	1.40	498	0,10%	697	8.58	12.99	55.80	54.60		8.16	±	0.92
LEADER(72)	GLP1A	Placebo	3.80	9340	4,92%	35492	34.28	39.09	36.00	64.40	12.90	8.70	±	1.50
LIRA-RENAL(73)	GLP1A	Placebo	0.50	277	0,02%	139	57.14	14.60	52.60	66.30	14.20	8.00	±	0.85
LIRA-SWITCH(74)	GLP1A	DPP4i	0.50	406	0,03%	203	0	9.80	39.00	56.50	7.60	8.20		
Laakso (2015) (75)	DPP4i	Sulfonylurea	1.00	241	0,03%	241	8.47	48.78	35.20	65.90		8.03	±	0.94
Leiter (2014) (76)	SGLT2i	Placebo	0.96	962	0,13%	924	13.02	12.97	33.00	63.60	13.00	8.10	±	0.80
Leiter (2015) (77)	SGLT2i	Sulfonylurea	2.00	1450	0,40%	2900	5.68	5.19	45.40	56.30	6.60	7.80	±	0.80
Lewin (2015) (78)	SGLT2i	DPP4i	1.00	398	0,06%	398	3.77	15.04	43.60	53.80		8.05	±	0.89
Lewin (2015)-1(78)	SGLT2i+DPP4i	DPP4i	1.00	402	0,06%	402	7.43	15.04	43.60	53.80		8.05	±	0.89
Lewin (2015)-2(78)	SGLT2i	DPP4i	1.00	398	0,06%	398	3.77	15.04	43.60	53.80		8.05	±	0.89
Mathieu (2016) (79)	SGLT2i+DPP4i	Placebo	1.00	320	0,04%	320	6.25	0	52.50	55.00	8.00	8.17	±	0.98
Matthews (2010) (80)	DPP4i	Sulfonylurea	0.96	3118	0,41%	2993	10	8.70	46.10	57.50	5.70	7.30	±	0.70

Matthews (2019) (81)	DPP4i	Placebo	5.00	2001	1,39%	10005	4.81	6.58	51.30	54.60	0.28	6.70	±	0.50
Mazzone (2006) (82)	Pioglitazone	Sulfonylurea	1.50	458	0,10%	687	0	5.85	37.30	59.90	7.50	7.40	±	0.97
Merker (2015) (83)	SGLT2i	Placebo	1.40	637	0,12%	892	4.98	13.80	44.00	56.00		7.90	±	0.70
Moses (2015) (84)	DPP4i	SGLT2i	1.00	422	0,06%	422	4.76	0	53.80	55.40	8.00	8.40	±	0.90
Muller-Wieland (2018)-1(85)	SGLT2i	Sulfonylurea	0.96	627	0,08%	602	6.63	3.33	33.50	58.60	6.70	8.30	±	0.80
Muller-Wieland (2018)-2(85)	SGLT2i+DPP4i	Sulfonylurea	0.96	625	0,08%	600	3.34	3.33	33.50	58.60	6.70	8.30	±	0.80
Muller-Wieland (2018)-3(85)	SGLT2i	Sulfonylurea	0.96	939	0,12%	901	6.66	0	33.50	58.60	6.70	8.30	±	0.80
Nauck (2016)-1(86)	DPP4i+TZD	Pioglitazone	0.63	801	0,07%	501	8.16	11.74	47.00	56.60		8.13	±	0.90
Nauck (2016)-2(86)	DPP4i+TZD	DPP4i	0.63	527	0,05%	329	8.16	23.70	38.00	56.00		8.00	±	0.90
Nissen (2008) (87)	Pioglitazone	Sulfonylurea	1.50	543	0,11%	815	12.35	14.65	34.10	59.70		7.40	±	1.00
Olansky (2011) (88)	DPP4i	Metformin	0.81	1246	0,14%	1009	1.98	0	43.00	50	3.20	9.80	±	1.80
PIONEER-3(89)	GLP1A	DPP4i	1.40	1863	0,36%	2608	8.19	4.59	49.00	58.00	8.80	8.30	±	0.90
PIONEER-6(90)	GLP1A	Placebo	1.30	3183	0,57%	4138	29.49	36.72	31.40	66.00	15.10	8.20	±	1.60
PIONEER-8(91)	GLP1A	Placebo	0.96	731	0,10%	702	17.14	22.64	42.90	60	14.80	8.20	±	0.70
Perez (2010)-2 (92)	Pioglitazone	Metformin	0.50	399	0,03%	200	10.58	0	53.00	53.70		8.65	±	0.07

Perkovic (2019) (93)	SGLT2i	Placebo	2.62	4401	1,60%	11531	37.61	46.69	33.30	63.20	16.00	8.30	±	1.30
Pfutzner (2011-A) (94)	DPP4i	Placebo	1.40	1304	0,25%	1826	5.12	6.53	50.30	51.80	1.70	9.40	±	1.30
Pinget (2013) (95)	GLP1A	Placebo	1.40	484	0,09%	678	0	8.87	49.00	55.30	8.10	8.10	±	0.80
Pratley (2018) (96)	SGLT2i	DPP4i	1.00	745	0,10%	745	2.01	0	38.70	54.80	6.20	8.50	±	1.00
REWIND(97)	GLP1A	Placebo	5.40	9901	7,41%	53465	22.23	24.79	46.10	66.20	10.60	7.40	±	1.10
Ridderstrale (2018) (98)	SGLT2i	Sulfonylurea	4.00	1545	0,86%	6180	4.90	8.01	46.00	55.70		7.92	±	0.86
Roden (2015)-1 (99)	SGLT2i	DPP4i	1.50	671	0,14%	1007	7.44	5.98	36.80	55.10		7.85	±	0.79
Roden (2015)-2 (99)	SGLT2i	Placebo	1.50	676	0,14%	1014	7.44	2.92	46.10	54.90		7.91	±	0.78
Rosenstock (2013) (100)	DPP4i	Sulfonylurea	1.00	441	0,06%	441	0	9.13	56.20	69.80	5.94	7.45	±	0.63
Rosenstock (2014) (101)	SGLT2i	Placebo	1.00	563	0,08%	563	5.33	5.32	60	55.30		8.33	±	7.90
Rosenstock (2015)-1(102)	SGLT2i	Placebo	1.50	494	0,10%	741	14.40	11.76	47.00	58.10		8.20	±	0.80
Rosenstock (2019)-1(103)	SGLT2i+DPP4i	SGLT2i	0.46	579	0,04%	266	0	7.52	47.40	55.90	7.60	8.20	±	0.90
SCORED (104)	SGLT2i	Placebo	2.75	10584	4,03%	29106	124.7	160.7	44.90	69		8.3		
SAVOR-TIMI 53(105)	DPP4i	Placebo	2.10	16492	4,80%	34633	35.25	35.31	32.70	65.00	10.30	8.00	±	1.40
SCALE(106)	GLP1A	Placebo	1.03	846	0,12%	871	7.66	13.74	54.20	54.70	6.70	7.90	±	0.80
SUSTAIN-1(107)	GLP1A	Placebo	0.57	387	0,03%	221	6.80	0	46.00	53.90	4.06	7.95	±	0.85

SUSTAIN-6(108)	GLP1A	Placebo	1.90	3297	0,87%	6264	34.49	46.60	41.50	64.80	14.00	8.70	±	1.49
SUSTAIN-8(109)	GLP1A	SGLT2i	0.96	788	0,10%	756	2.64	2.64	49.00	57.50	7.20	8.20	±	1.00
Schernthaner (2013) (110)	SGLT2i	DPP4i	0.96	755	0,10%	725	8.29	5.51	43.10	56.70	9.70	8.10	±	0.90
Schernthaner (2015) (111)	DPP4i	Sulfonylurea	0.96	720	0,10%	691	11.57	8.68	36.70	72.70	7.60	7.62	±	0.65
Schweizer (2009) (112)	DPP4i	Metformin	0.46	335	0,02%	154	12.86	13.10	47.00	70.20	3.00	7.70	±	0.60
Scott (2018) (113)	SGLT2i	SGLT2i	0.46	613	0,04%	282	0	7.08	45.00	67.70	10.50	7.70	±	0.70
Seck (2010) (114)	DPP4i	Sulfonylurea	2.00	1172	0,32%	2344	0.85	3.42	38.70	56.60	6.20	7.60	±	0.90
Seino (2010) (115)	GLP1A	Sulfonylurea	0.46	400	0,03%	184	24.33	49.41	35.00	58.50	8.50	8.78	±	0.97
Seino (2016) (116)	GLP1A+Insulin	Insulin+Placebo	0.69	257	0,02%	177	0	22.30	42.30	59.80	14.69	8.80	±	0.90
Soloist-WHF (2020) (117)	SGLT2i	Placebo	0.75	1222	0,13%	916			33.75	69.00		7.15		
Strojek (2014) (118)	SGLT2i	Placebo	0.90	592	0,07%	533	9.94	7.66	51.00	60.30	7.40	8.15	±	0.80
TECOS(119)	DPP4i	Placebo	3.00	14671	6,10%	44013	33.87	33.88	29.50	65.50	11.60	7.20	±	0.50
Tolman (2009) (120)	Pioglitazone	Sulfonylurea	3.00	2097	0,87%	6291	5.71	8.29	44.50	55.00	5.40	9.50	±	2.00
VADT (2009) (121)	Sulfonylurea+Insulin	Sulfonylurea+Insulin	5.60	1791	1,39%	10030	47.05	52.44	26.00	60.4	11.50	9.40		
Wainstein (2011) (122)	Pioglitazone	DPP4i+Metformin	0.61	517	0,04%	315	0	6.28	45.20	52.40	3.20	9.00	±	1.30
Wilding (2012) (123)	SGLT2i	Placebo	0.90	800	0,10%	720	3.66	0	50.80	58.80	13.50	8.47	±	0.77

Wilding (2013) (124)	SGLT2i	Placebo	1.00	469	0,07%	469	3.19	12.82	51.30	56.80	10.30	8.10	±	0.90
Yale (2014) (125)	SGLT2i	Placebo	1.00	269	0,04%	269	27.93	33.33	36.70	68.20	16.40	8.00	±	0.90
Yang (2018) (126)	SGLT2i	Placebo	0.46	272	0,02%	125	0	16.35	51.90	58.60	12.20	8.58	±	0.81

Continuous data are presented as mean ± SD.

RCTs in red the studies with at least 1% of overall weight, which is based on the ratio between each individual study exposure (in patient-years) vs overall exposure.

Most of large RCTs are noted as their acronyms and other studies are annotated as First author (publication year).

The presence of dash followed by a number (e.g. Rosenstock (2015)-1) denotes that one of multiple study arms is included.

This table excluded studies or study arms in which no clinical event occurred in the active or in the control arm.

Table S2. Risk of bias analysis

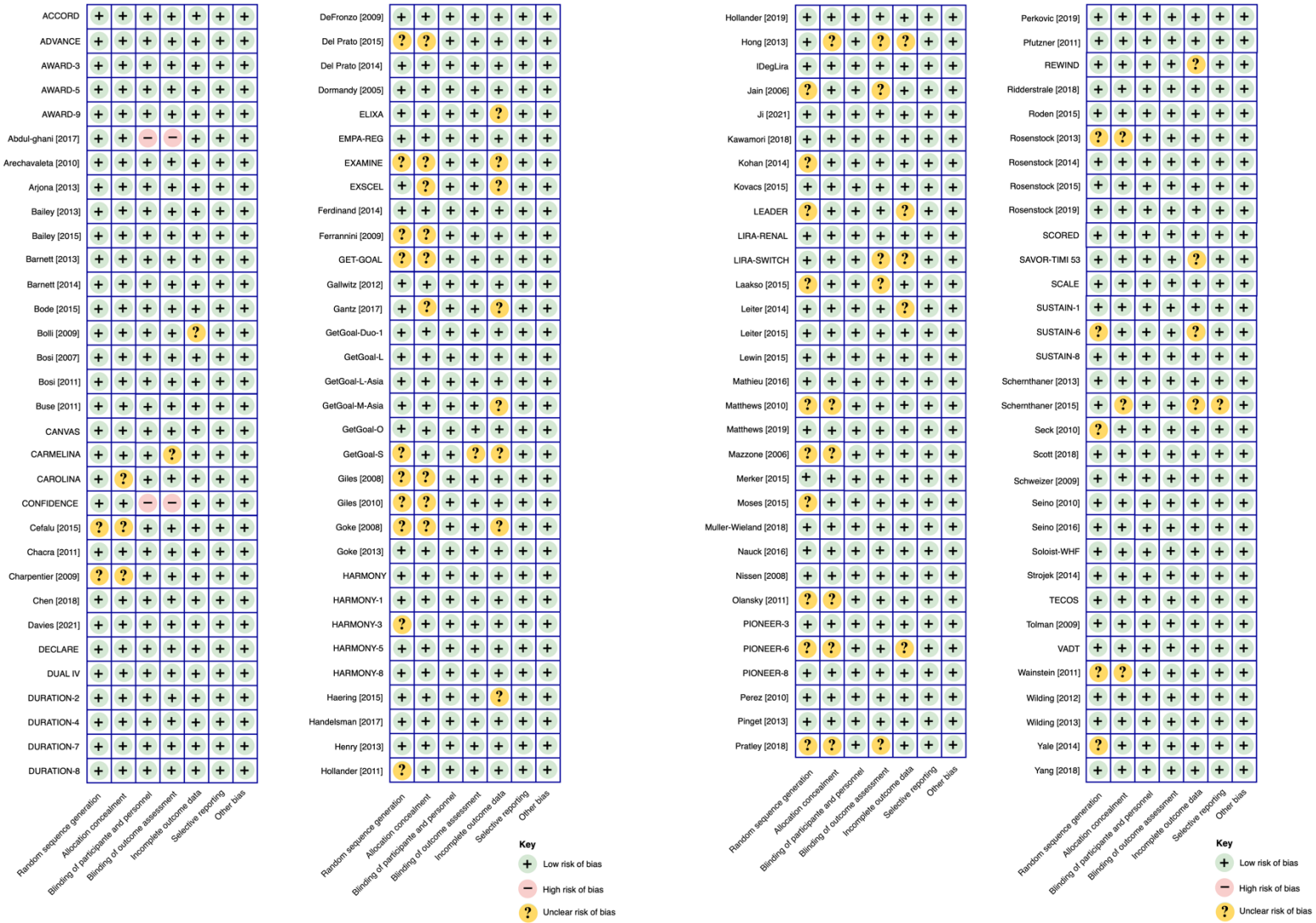
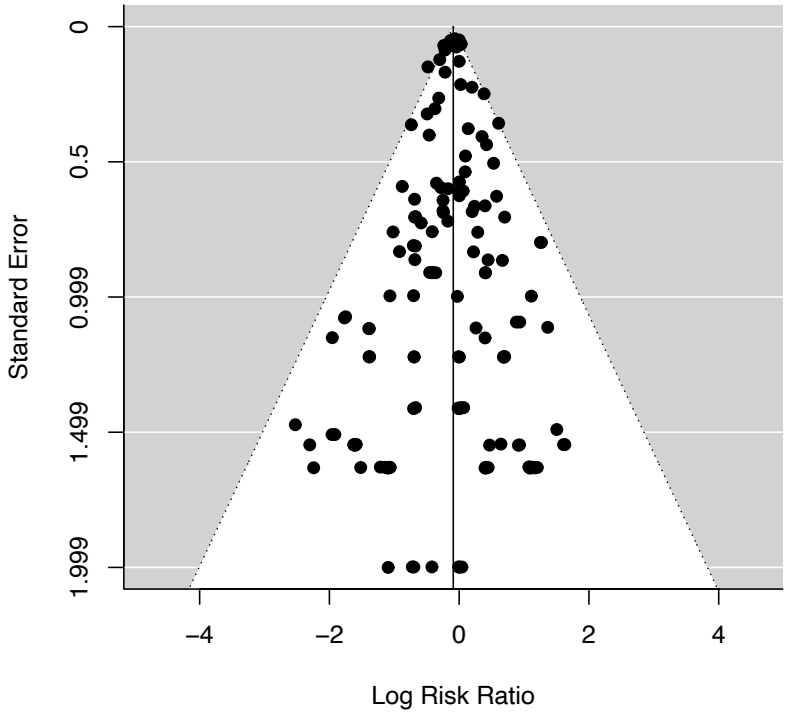


Table S3. Egger's Regression Tests for Funnel Plot Asymmetry

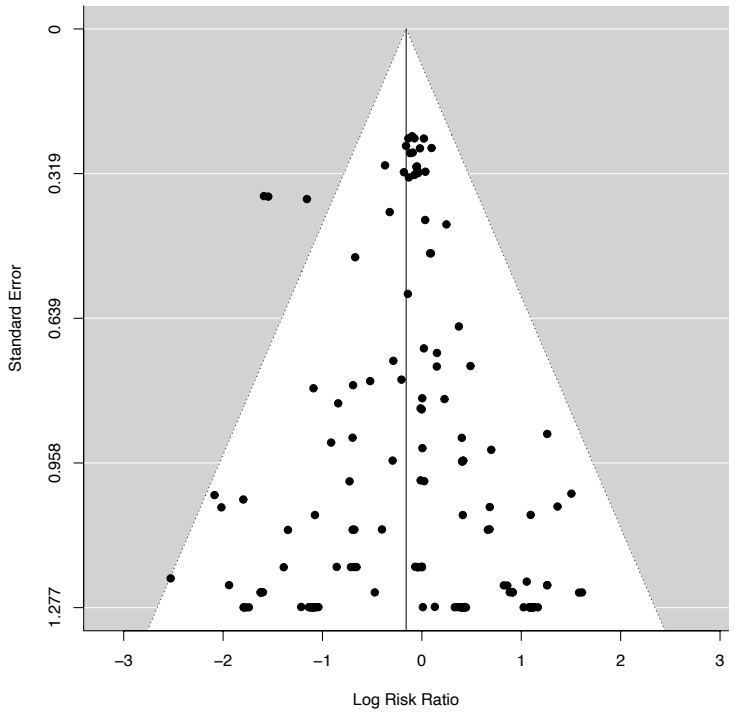
		t	z	p-value
MACE				
	weighted regression with multiplicative dispersion	-1.3508		0.1787
	mixed-effects meta-regression model		-1.0985	0.2720
Non-fatal MI				
	weighted regression with multiplicative dispersion	-0.9182		0.3598
	mixed-effects meta-regression model		0.0574	0.9542
All-cause death				
	weighted regression with multiplicative dispersion	0.5530		0.5811
	mixed-effects meta-regression model		0.5723	0.5671

Figure S1. Evaluation of publication bias in funnel plots for (a) MACE; (b) All-cause deaths; (c) Non-fatal myocardial infarction

(a)



(b)



(c)

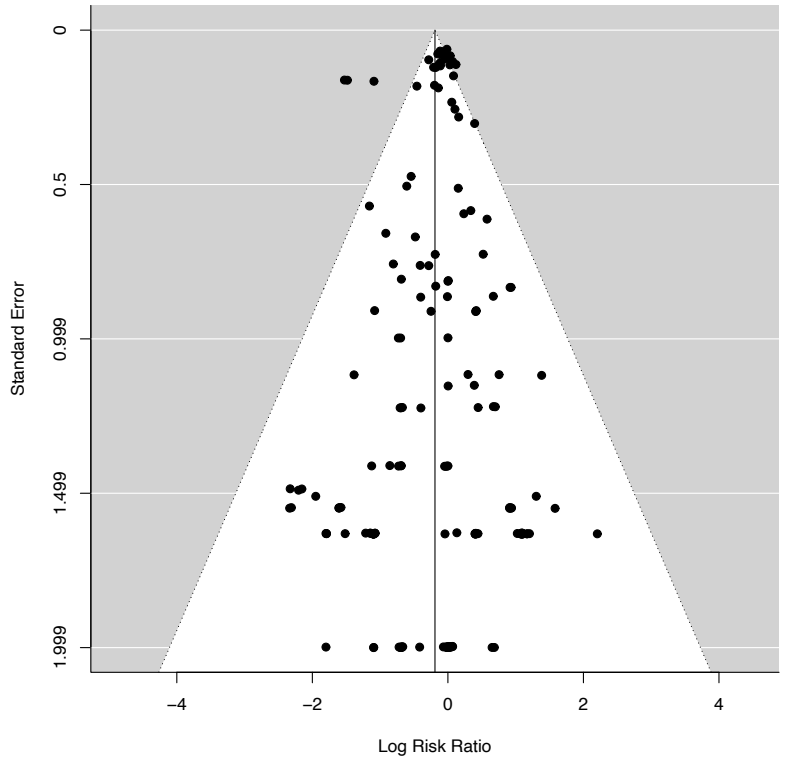
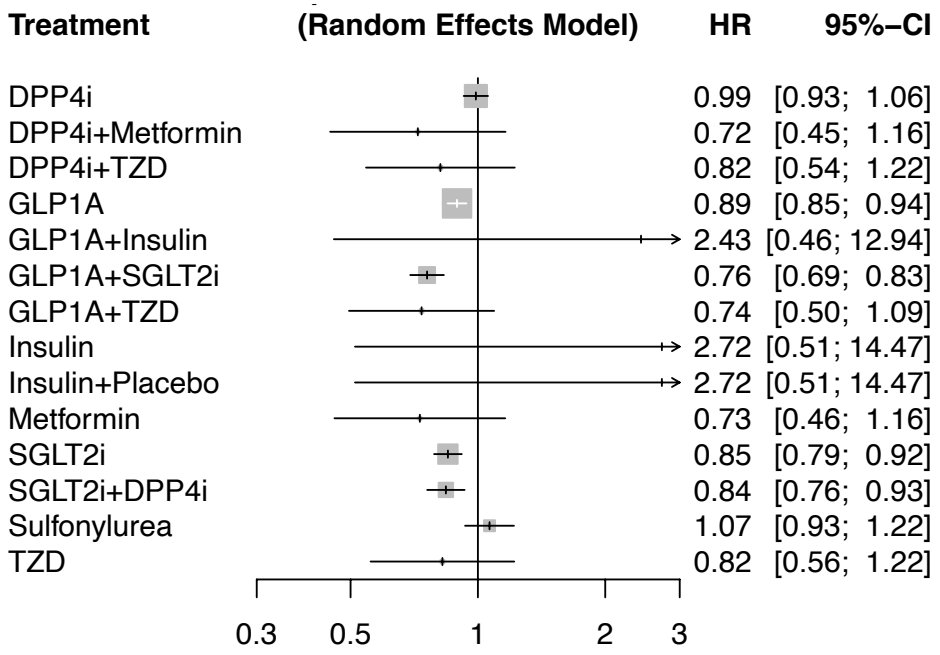


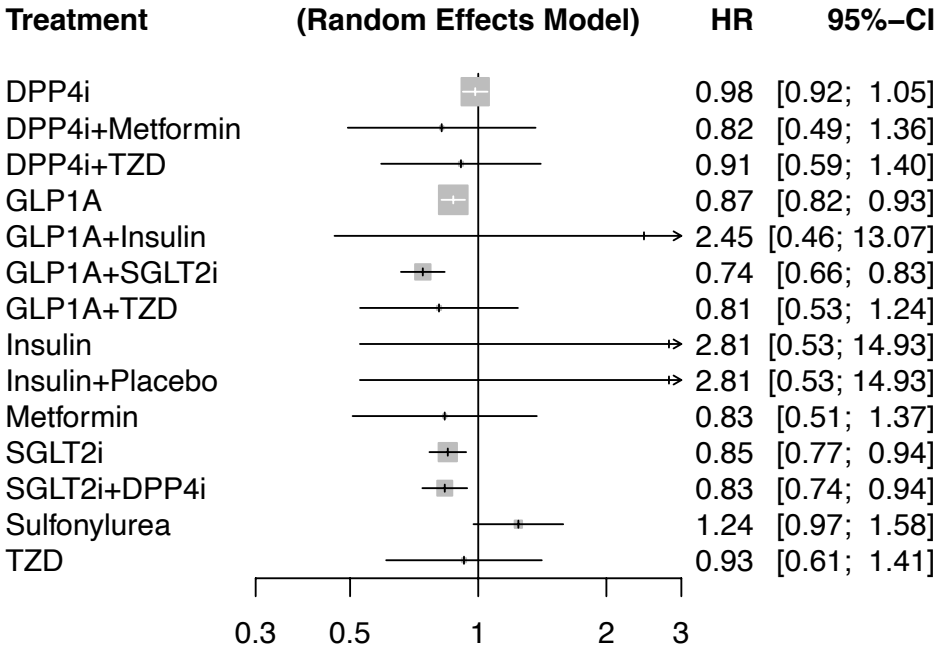
Figure S2. Sensitivity analysis with exclusion of one study with the largest exposure (sample size * follow-up time) per drug in the active arm. Forest plot comparing antidiabetic therapies for the occurrence of major cardiovascular adverse events (MACE) in an additive effects network meta-analysis with a random-effects model.

(a) Sensitivity analysis 1



The excluded trials were: TECOS, REWIND, DECLARE, Dormandy (2005). The reference group was considered as placebo. Number of pairwise comparisons was 131 (118 studies with >100 individuals per arm), number of treatments (n = 19), number of designs (n=37) with 7 active components. Heterogeneity / inconsistency analysis showed $tau^2 = 0$; tau = 0; $I^2 = 0\%$ [0.0%; 0.0%]; Q = 98.58 (p=0.9890). Among TZDs (thiazolidinediones) only pioglitazone was included.

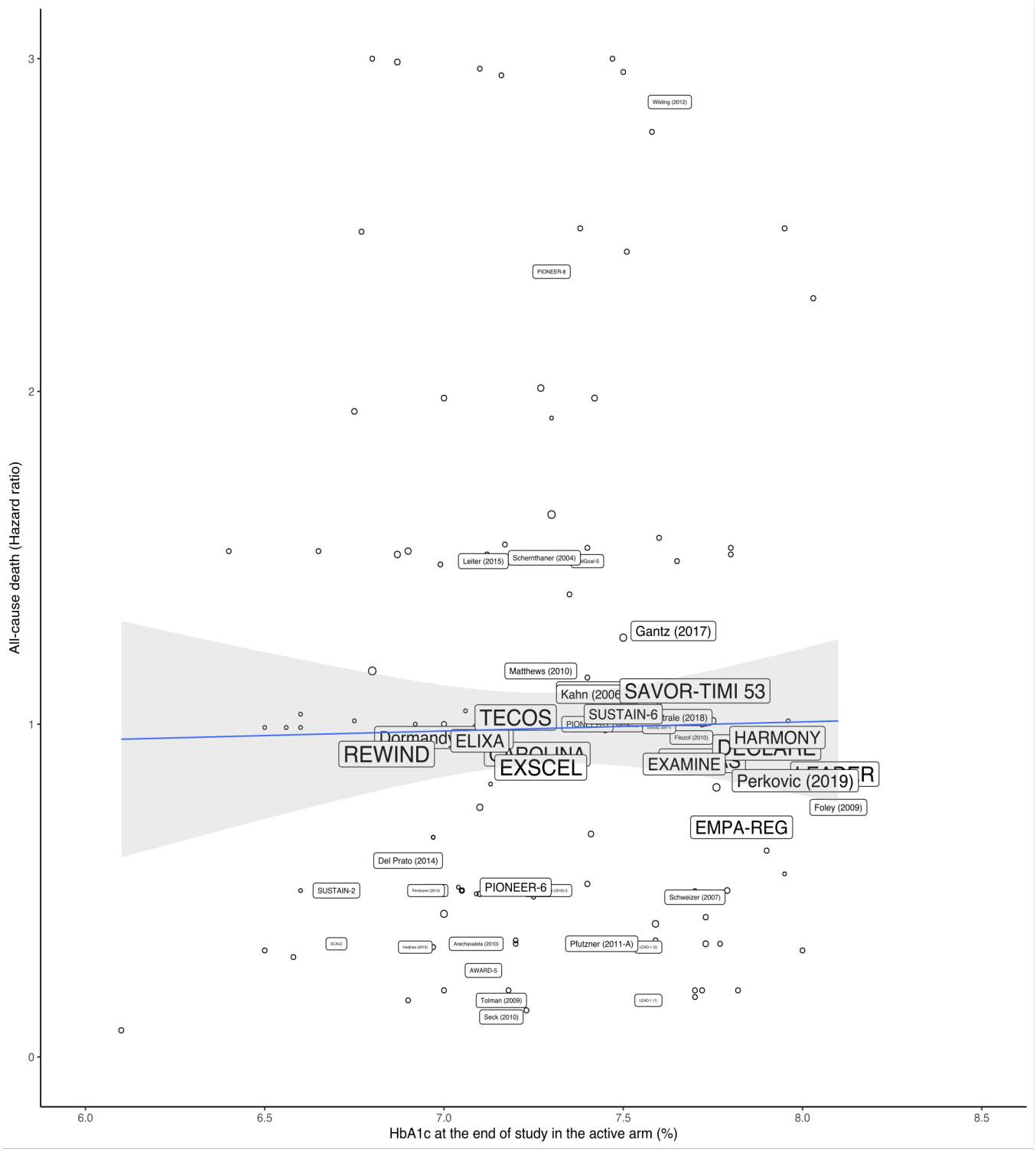
(b) Sensitivity analysis 2



The excluded trials were: TECOS, CAROLINA, REWIND, EXCEL, DECLARE, CANVAS, Dormandy (2005). Number of pairwise comparisons was 128 (115 studies with >100 individuals per arm), number of treatments (n = 19), number of designs (n=37) with 7 active components. Heterogeneity / inconsistency analysis showed $tau^2 = 0$; tau = 0; $I^2 = 0\%$ [0.0%; 0.0%]; Q = 84.01 (p=0.9912). Among TZDs (thiazolidinediones) only pioglitazone was included.

Figure S3. Sensitivity analysis considering (a) all-cause death or (b) non-fatal MI as outcomes

(a)



(b)

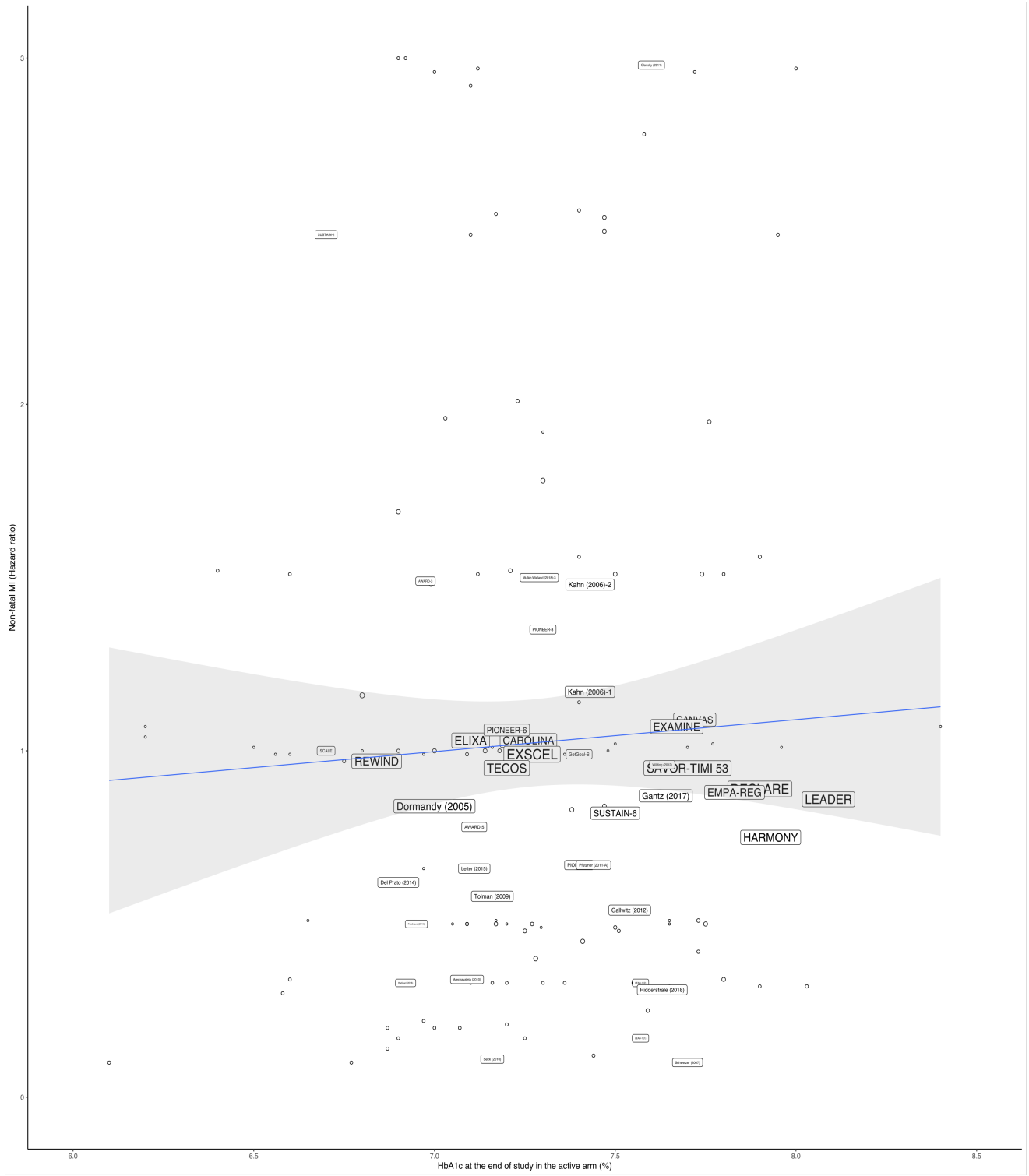

Non-fatal MI (Hazard ratio)
HbA1c at the end of study in the active arm (%)
0
1
2
3
6.0
6.5
7.0
7.5
8.0
8.5
Kahn (2006)-2
Kahn (2006)-1
PIONEER-6
ELIXA
CAROLINA
EXSCEL
REWIND
TECOS
EXAMINE
Gantz (2017)
SUSTAIN-6
LEADER
HARMONY
Dormandy (2005)
Leiter (2015)
Del Prato (2014)
Tolman (2009)
Gallwitz (2012)
Ridderstrale (2018)

Figure S4. Relationship between weight change and the HR for MACE

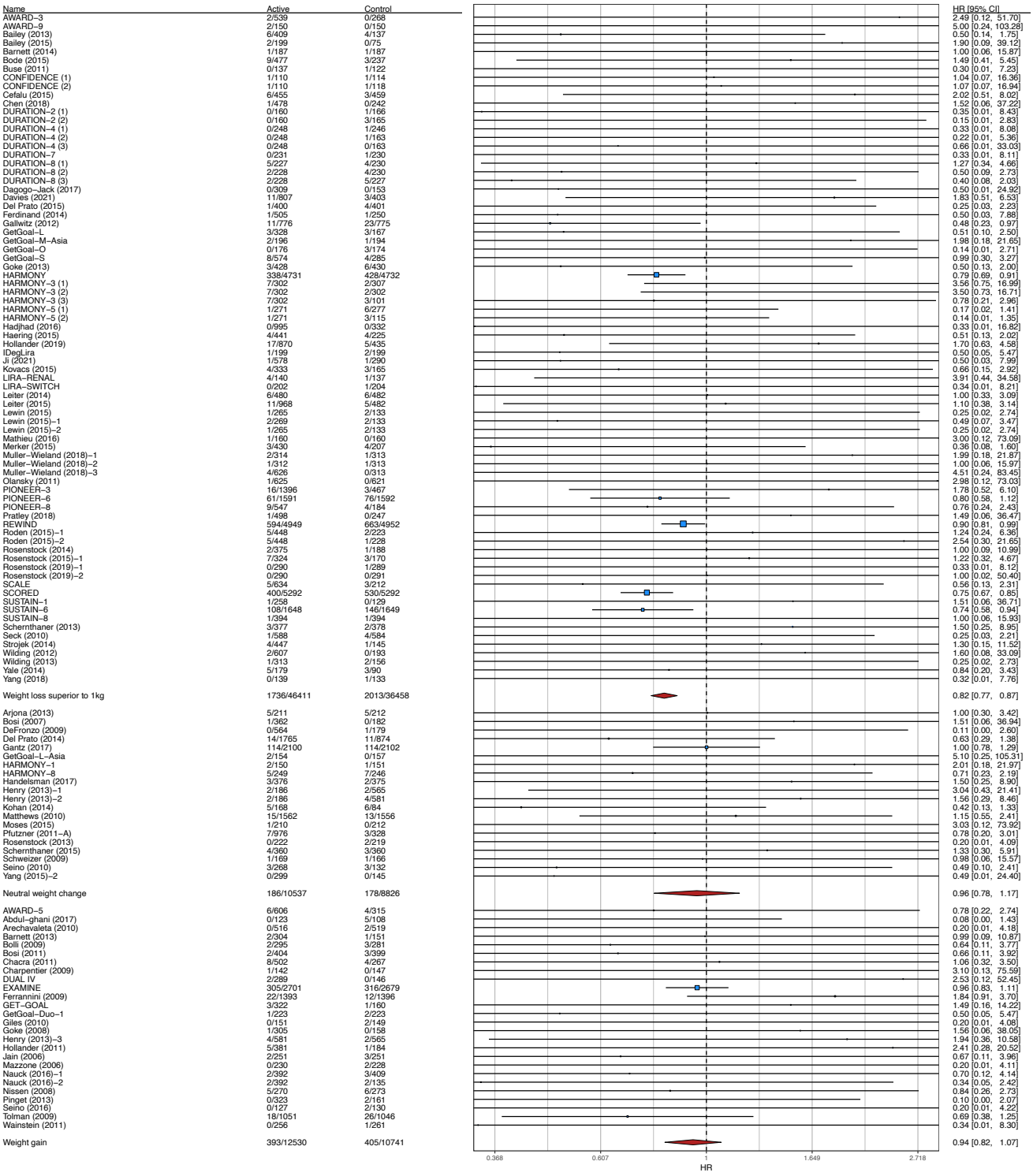
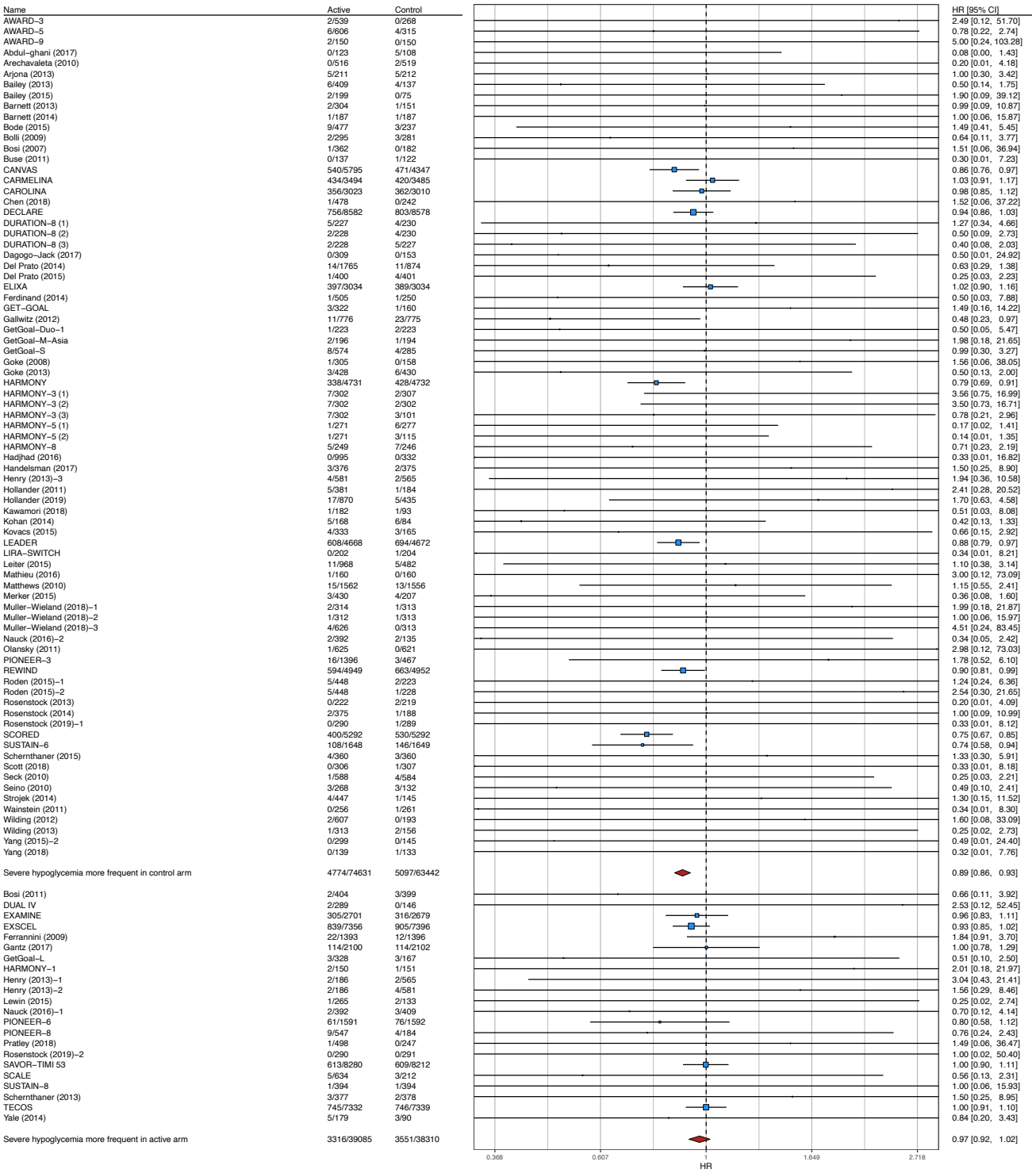


Figure S5. Relationship between the occurrence of severe hypoglycemia and the HR for MACE



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